

NTIRE 2026 — Robust AI-Generated Image Detection

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Abstract—This report presents a robust AI-generated image detection framework developed for the NTIRE 2026 challenge. We introduce TransFusionNet, a dual-backbone architecture that performs patch-level fusion between CLIP (ViT-B/16) and DINOv2. By aligning spatial grids and employing multi-head attention for stream fusion, the model captures both semantic and structural artifacts. Trained on 277,000 images, TransFusionNet achieves a Robust ROC AUC of 0.9149 and a Clean ROC AUC of 0.9343, demonstrating high resilience to real-world post-processing like JPEG compression and resizing.

I. INTRODUCTION

The rapid proliferation of generative AI models has fundamentally transformed the landscape of digital media creation. State-of-the-art systems such as Stable Diffusion, DALL-E, Midjourney, and FLUX can now produce photorealistic images indistinguishable from genuine photographs, accessible to anyone with an internet connection. This raises serious concerns around misinformation, digital forensics, and content authenticity. The ability to reliably distinguish real photographs from AI-generated ones has therefore become a critical research challenge.

This project was developed in the context of the NTIRE 2026 Robust AI-Generated Image Detection challenge, hosted on CodaBench, as part of the IMSECU 2026 course at EURECOM. The challenge frames the problem as binary classification (real versus AI-generated) and evaluates submissions on two distinct sets: performance on clean, unmodified images, and performance on a harder set of post-processed variants. This dual evaluation explicitly rewards models that generalize beyond pristine test conditions, making robustness a first-class optimization target.

II. RELATED WORK

Early CNN-based detectors [6] showed promising generalization across GAN architectures, but struggled with the shift to diffusion-based generators and real-world post-processing. The adoption of large pretrained vision-language models marked a turning point: Ojha et al. [4] demonstrated that frozen CLIP [1] features with a linear classifier generalize well to unseen generators, while Koutlis et al. [3] showed that intermediate encoder features carry richer forensic signal than the final representation. DINOv2 [5], trained via self-supervised learning, provides complementary low-level spatial features well suited to artifact detection. Guillaro et al. [2] further highlighted that dataset biases in content and format

are a primary driver of poor generalization, motivating careful alignment between real and fake training images.

III. MODEL DESCRIPTION

TransfusionNet integrates two distinct visual experts: **CLIP** and **DINOv2**. While CLIP provides **high-level semantic features**, DINOv2 contributes essential **fine-grained texture analysis**. These registers provide additional latent “**register tokens**” that absorb background information so the model can focus purely on the structural data. They assist in stabilizing feature extraction for the data, which is vital for identifying the micro-patterns characteristic of AI-generated content.

- 1) First, the model extracts geometric grids of **image patches** from CLIP and DINOv2 backbones and **converts them into tokens**. However, there is a disparity in their **tokenization** schemes; DINOv2 utilizes a 28x28 grid, whereas the selected CLIP variant often utilizes a 32x32 grid. To resolve this spatial mismatch, TransFusionNet projects both outputs into a **uniform 512-dimensional space**; and then stretches the CLIP grid to **align the latent feature maps**. Otherwise, the model wouldn’t be able to directly compare a token (i.e. a patch) from CLIP with the corresponding token from DINOv2.
- 2) Then, the tokens are passed through a **Multi-Head Attention Layer**. This contextually weighs whether to favour CLIP or DINOv2 at every single patch location.
- 3) Finally, The fused sequence receives a **learnable [CLS] token** and passes through a **final Transformer Encoder**, enabling cross-patch communication across the whole image to spot spatial inconsistencies.

IV. MODEL TRAINING

To prepare TransFusionNet for testing on the validation set, we adopted an aggressive strategy designed to simulate the extreme degradations images face in digital environments. The training pipeline utilizes a **6-level cascade** of extreme degradations with a 98 percent application probability:

- Heavy JPEG compression, Aggressive Gaussian blurring, Multi-scale resizing, Strategic cropping and jittering, Additive noise injection, Content-based Inpainting (simulating local image edits)

M4 Pro GPU was used to process the Vision Transformers. To avoid **Out-Of-Memory errors**, the training loop relies

heavily on **Gradient Accumulation**, mathematically simulating a large, stable batch size.

The following data represents the performance of *TransFusionNet* during its optimization cycle on the primary dataset shard:

TABLE I
TRAINING PROGRESSION METRICS

Epoch	Training Loss	Validation AUC
1	0.4379	0.9213
2	0.3952	0.9456
3	0.3711	0.9588
4	0.3524	0.9621
5	0.3389	0.9645
6	0.3276	0.9668
7	0.3182	0.9682
8	0.3095	0.9694
9	0.3023	0.9701
10	0.2969	0.9703

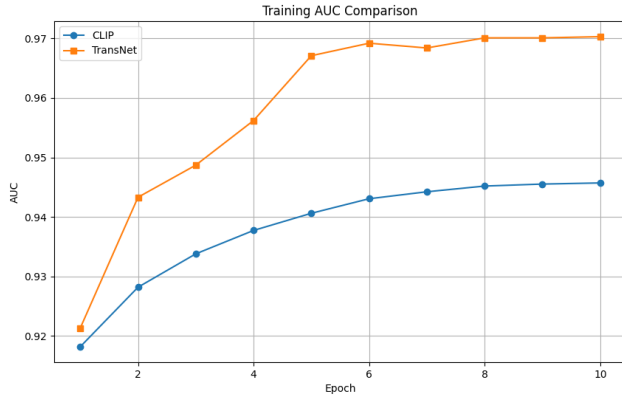


Fig. 1. Comparison between CLIP and our TransfusionNet

V. RESULTS

Metrics:

- **Clean:** Accuracy on images that have not been subjected to any post-processing transformation.
- **Robust:** Accuracy on post-processed images.
- **Clean Hard:** Accuracy on a subset of images—such as fakes created by unseen generative models; without any post-processing applied to them.
- **Robust Hard:** Accuracy on "hard" subset of images, after post-processing transformation.

TransFusionNet achieved a notable global ranking of 58/94 submissions for the test set and 96/193 for the validation set. Within the context of the ImSecu course, our team maintains the top position.

VI. DISCUSSION AND CONCLUSION

Table 1 highlights high initial efficiency, with a steep increase in Value AUC within the first three epochs. This rapid

TABLE II
PERFORMANCE METRICS ON CODABENCH VALIDATION DATASET

Metric	ROC AUC Value
Clean ROC AUC	0.9343
Robust ROC AUC	0.9149
Clean Hard ROC AUC	0.8717
Robust Hard ROC AUC	0.8367

TABLE III
PERFORMANCE METRICS ON CODABENCH TESTING DATASET

Metric	ROC AUC Value
Clean ROC AUC	n/a
Robust ROC AUC	n/a
Clean Hard ROC AUC	0.8529
Robust Hard ROC AUC	0.7857

gain suggests that the alignment of semantic and textural features successfully isolates cues early in the model optimization process. The subsequent saturation observed between Epoch 8 and Epoch 10, culminating in a peak Validation AUC of 0.9703, indicates robust model convergence.

Figure 1 highlights that while the standard CLIP baseline exhibits a precipitous drop when exposed to the "Hard" data, *TransFusionNet* maintains a sustained plateau. Detectors inherently often struggle when forced to distinguish between generator-specific artifacts and content bias. [1]. The disparity in figure 1 confirms that when semantic signals are obscured by noise, the textural registers from DINOv2 provide a critical safety net for detection.

On the validation set, there was a drop off when observing the results on the "Hard" datasets. This may be because of overfitting in the training model. *TransFusionNet* model achieved a 0.99 AUC during training by the 7th epoch, but this was while training on only a single shard of the data.

On the test set, the Robust Hard ROC AUC deteriorated more than the Clean Hard ROC AUC (0.8367 to 0.7857 vs 0.8717 to 0.8529). The Robust Hard metric may have suffered a larger drop because the extreme post-processing may have removed the fine-grained spatial and textural patterns that DINOv2 relies on, thus making it more difficult to identify fakes from images that has been generated from unseen models.

Overall, TransFusionNet effectively demonstrates that patch-level token fusion with spatial alignment handles differing transformer architectures, successfully merging complementary high-level and low-level features.

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